

Connecting macOS VS Code to SQL Server in a VMware Fusion Windows VM

A practical, repeatable setup guide based on a verified MacBook Pro -> Windows 11 VM -> SQL Server connection

MacBook Pro VS Code + mssql	VMware Fusion NAT / Share with my Mac	Windows 11 VM Firewall + TCP/IP	SQL Server TCP 1433
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Privacy note: All reused screenshots in this guide were sanitized for public sharing. Personal names, machine-specific identifiers, and network values were blurred where appropriate. Use your own VM IP address and credentials.

Verified workflow: VMware NAT -> SQL Server TCP/IP -> Windows Firewall -> Mixed Mode -> VS Code mssql

1. What this guide accomplishes

The goal is to run SQL Server inside a Windows virtual machine while writing and executing SQL from VS Code on the macOS host. The Mac talks directly to the VM over VMware Fusion's private NAT network.

- Host: macOS on a MacBook Pro.
- Hypervisor: VMware Fusion.
- Guest: Windows 11 Enterprise.
- Database: SQL Server default instance (MSSQLSERVER).
- Client: VS Code on macOS with Microsoft SQL Server (mssql).
- Transport: TCP port 1433 between the Mac host and Windows VM.

Connection path

1	VMware Fusion gives the Windows VM an address on a NAT network.
2	SQL Server listens on TCP/IP, port 1433.
3	Windows Firewall allows inbound TCP 1433.
4	The Mac confirms the port with nc.
5	VS Code connects using SQL Login authentication.

Values you will need

Setting	Value
VM IP	<VM_IP> from ipconfig
SQL TCP port	1433
Authentication	SQL Login / Mixed Mode
VS Code extension	SQL Server (mssql) by Microsoft

Important: The VM IP can change if VMware DHCP assigns a new address. If a connection stops working after a VM/network restart, run ipconfig again before changing other settings.

2. VMware Fusion networking

Step 1 - Open the VM Network Adapter settings

Shut down or safely pause the VM if VMware requires it for network changes. Open the VM settings and choose Network Adapter.



Figure 1 - VMware Fusion VM settings: open Network Adapter.

Step 2 - Use "Share with my Mac" (NAT)

Keep Connect Network Adapter enabled and select Share with my Mac. This is VMware NAT mode: the VM has its own private address while sharing the Mac's external network connectivity.

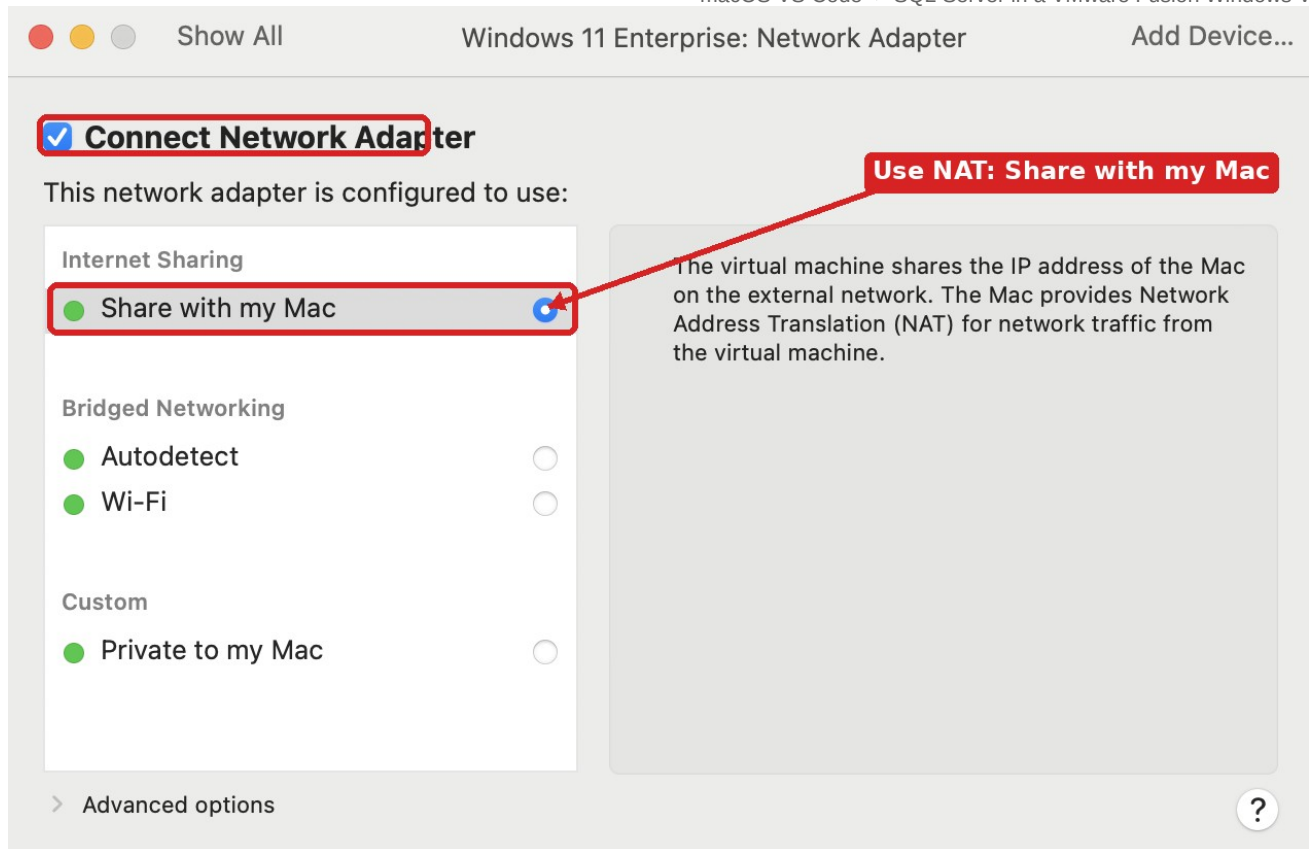


Figure 2 - Recommended VMware networking for this setup: Share with my Mac (NAT).

Why NAT? It is simple for a development VM: the VM can reach the internet, and the Mac host can reach the VM on the VMware private network. Bridged networking is not required for this host-to-VM SQL workflow.

3. Verify SQL Server and identify the VM address

Step 3 - Confirm SQL Server works inside Windows

Before troubleshooting networking, verify that SQL Server itself is running and that SSMS can connect locally to localhost.

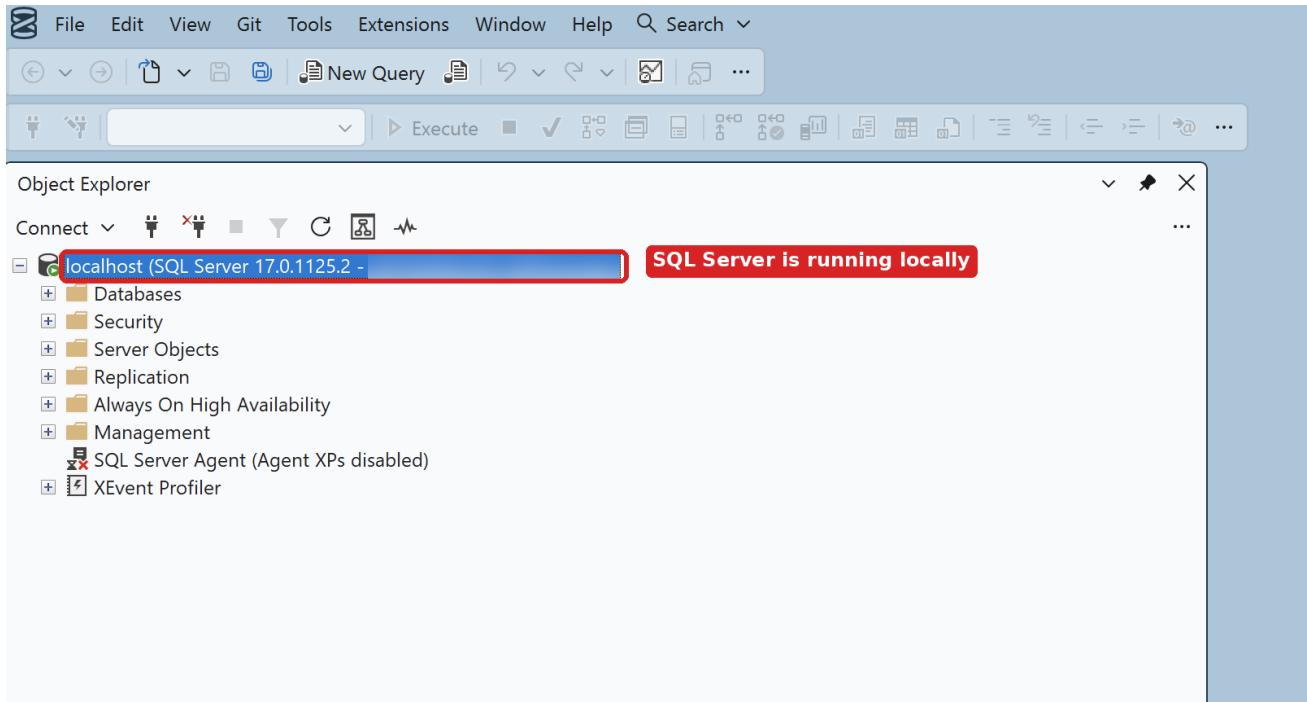


Figure 3 - SQL Server reachable locally inside the Windows VM. Machine/user details are blurred.

Step 4 - Find the VM IPv4 address

Open Command Prompt inside the Windows VM and run:

```
ipconfig
```

Under the active Ethernet adapter, find IPv4 Address. Treat that value as <VM_IP> for all later commands and connection settings.

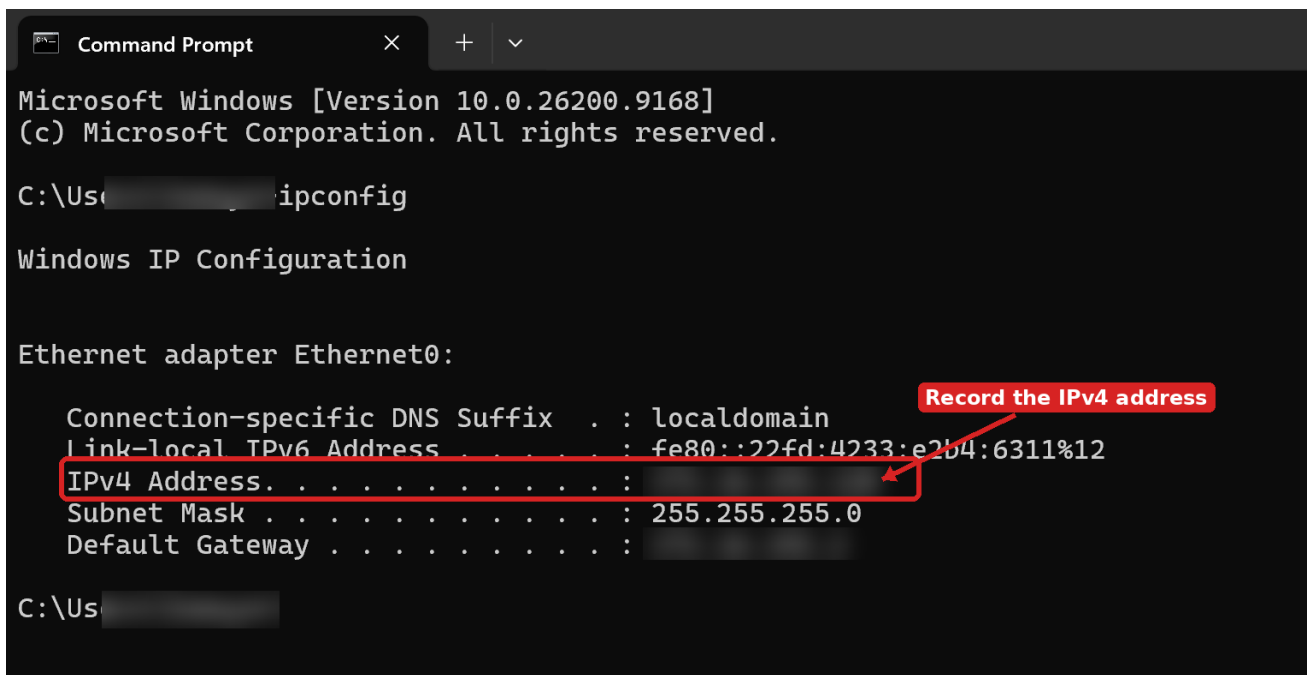


Figure 4 - Locate the IPv4 Address for the active VMware Ethernet adapter. User and network values are sanitized.

4. Enable SQL Server TCP/IP

Step 5 - Enable TCP/IP in SQL Server Configuration Manager

Inside Windows, open SQL Server Configuration Manager. Go to SQL Server Network Configuration -> Protocols for MSSQLSERVER. If TCP/IP is Disabled, right-click it and choose Enable.

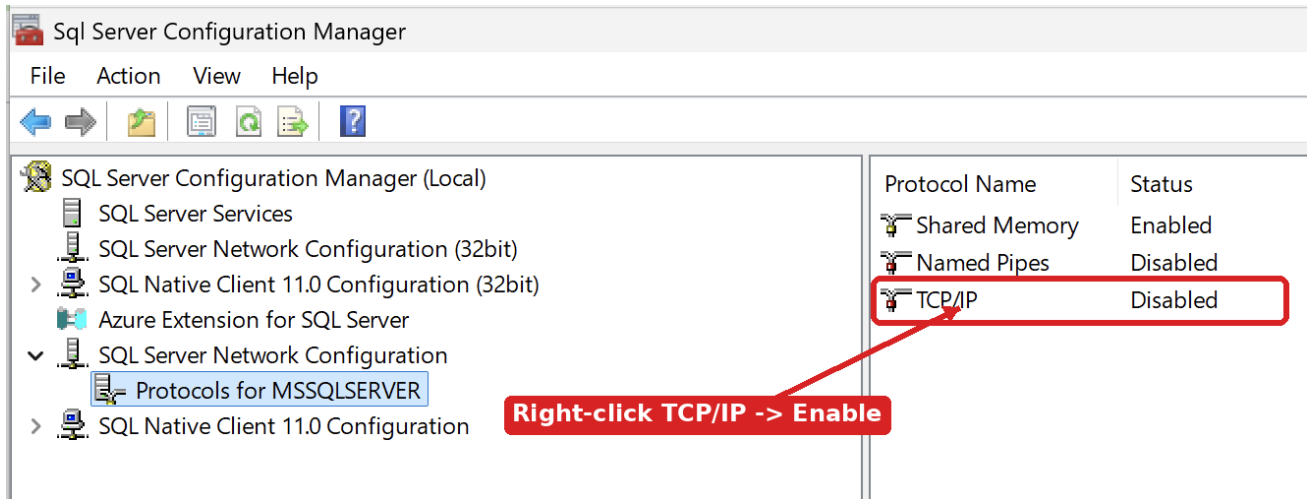


Figure 5 - TCP/IP must be enabled for remote TCP connections.

Step 6 - Restart the SQL Server service

After enabling TCP/IP, open SQL Server Services in Configuration Manager and restart SQL Server (MSSQLSERVER). Protocol changes do not take effect until the service restarts.

Step 7 - Confirm a fixed TCP port

Open TCP/IP Properties -> IP Addresses and scroll to IPAll. For this setup:

- TCP Dynamic Ports: leave empty.
- TCP Port: 1433.

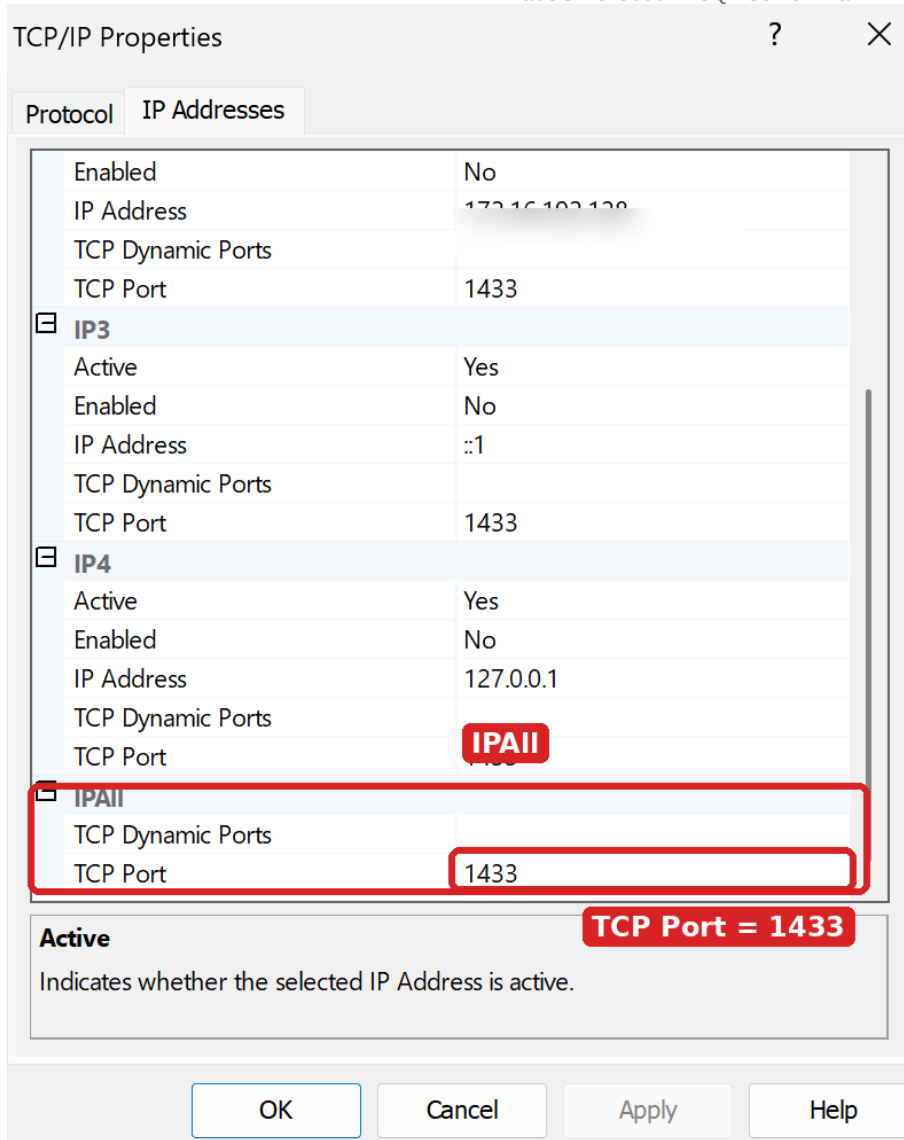


Figure 6 - IPAll uses fixed TCP port 1433; the VM IP shown in the original screenshot is blurred.

If you change the port: Restart SQL Server again after applying the change. If you use a different port, use that same port in the Windows Firewall rule, nc test, and VS Code connection.

5. Allow SQL Server through Windows Firewall

Step 8 - Open Windows Defender Firewall with Advanced Security

Press Win + R, run wf.msc, then select Inbound Rules.

wf.msc

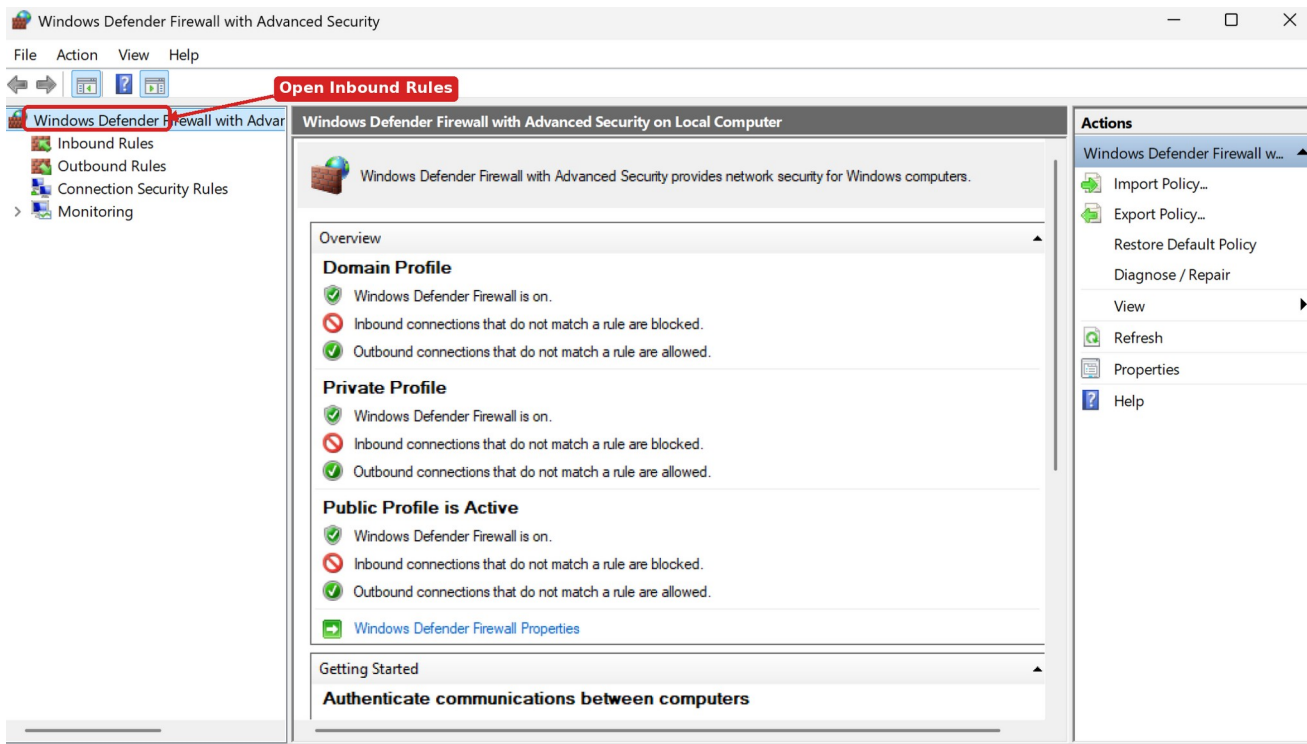


Figure 7 - Open Inbound Rules.

Step 9 - Create a Port rule

On the right, choose New Rule..., then select Port as the rule type.

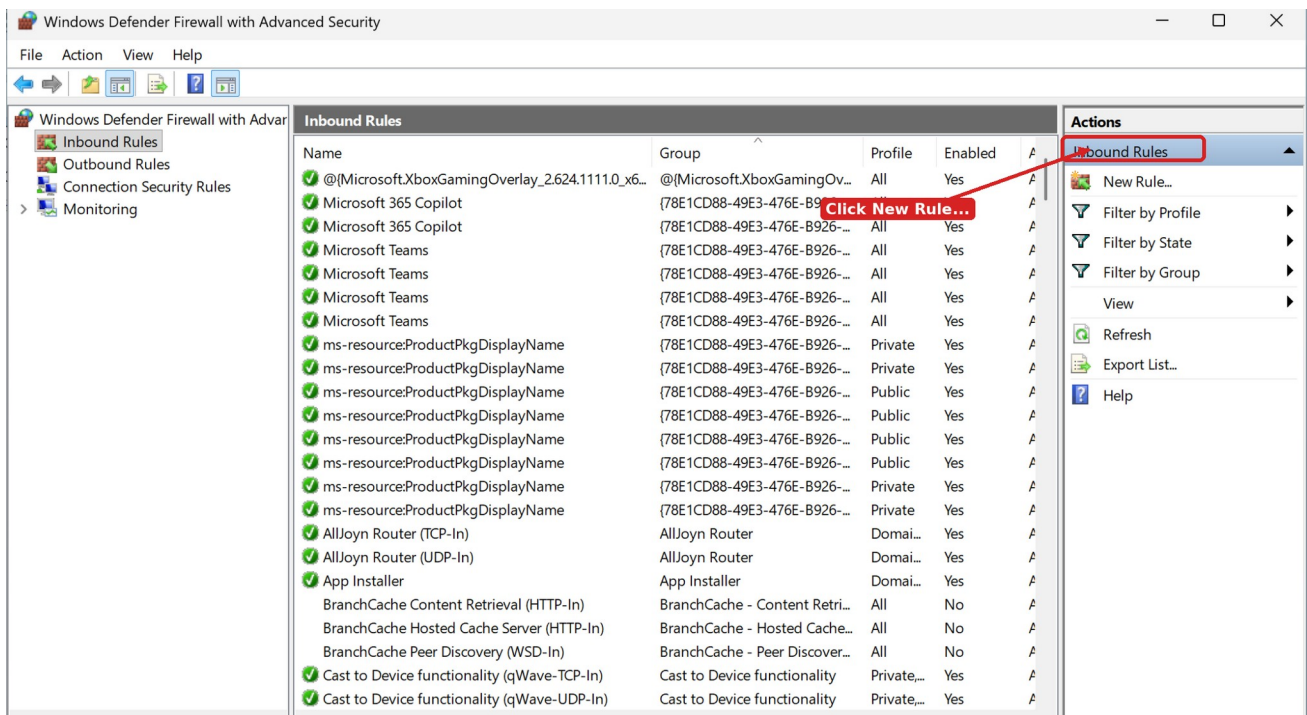


Figure 8 - Start a new inbound firewall rule.

6. Configure the inbound TCP 1433 rule

Step 10 - Protocol and port

Select TCP and Specific local ports, then enter 1433.

New Inbound Rule Wizard

Protocol and Ports

Specify the protocols and ports to which this rule applies.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

Does this rule apply to TCP or UDP?

☒ TCP

☐ UDP

Does this rule apply to all local ports or specific local ports?

☐ All local ports

☒ Specific local ports:

Example: 80, 443, 5000-5010

Specific local port: 1433

Figure 9 - Inbound rule for TCP port 1433.

Step 11 - Allow the connection and choose profiles

Choose Allow the connection. In the verified VM, Windows classified the VMware network as Public, so the rule was enabled for Private and Public. Domain was not needed.

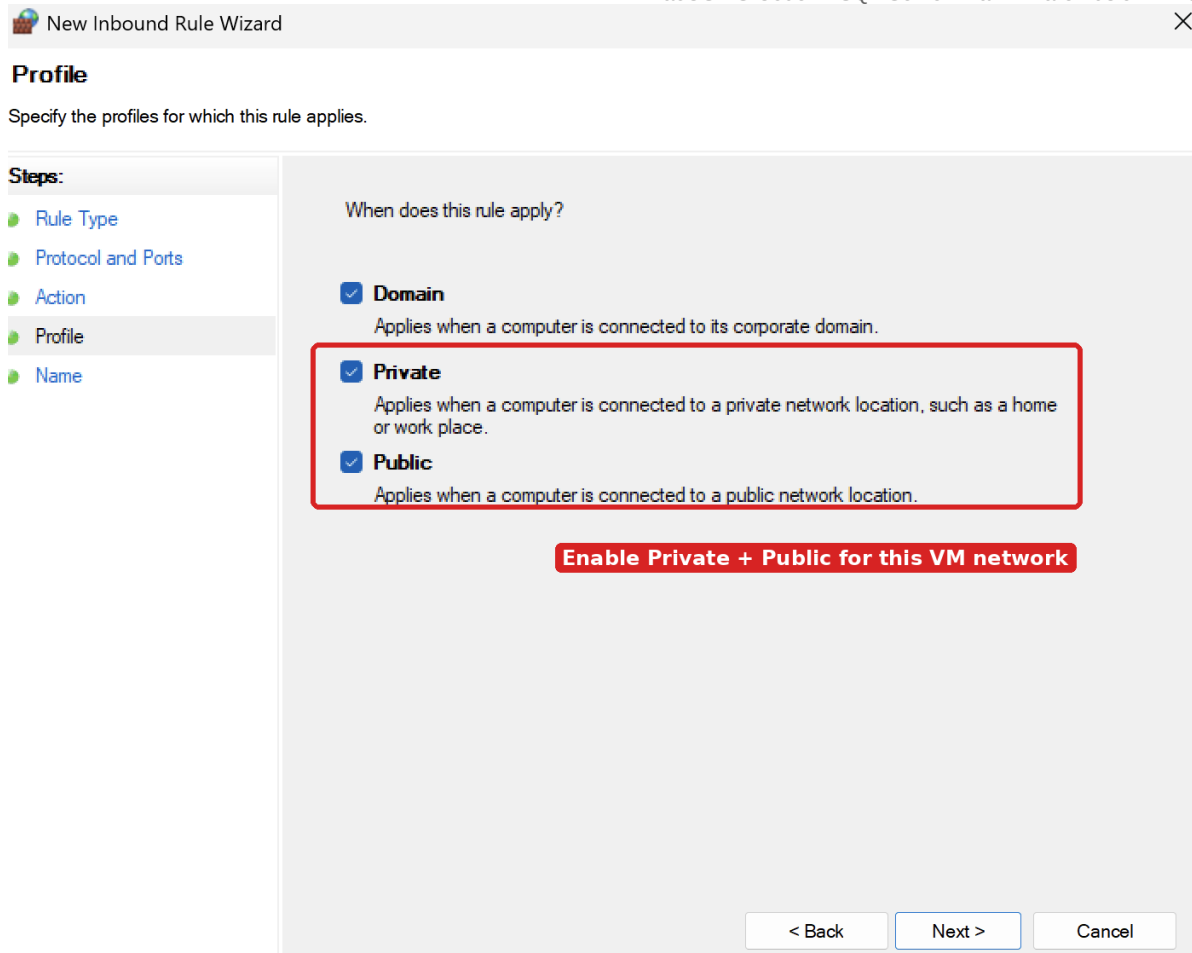


Figure 10 - Private + Public profiles cover the VMware VM network used in this setup.

Step 12 - Name and finish the rule

A clear name such as SQL-Server makes the rule easy to find later. Finish the wizard and confirm the rule is enabled.

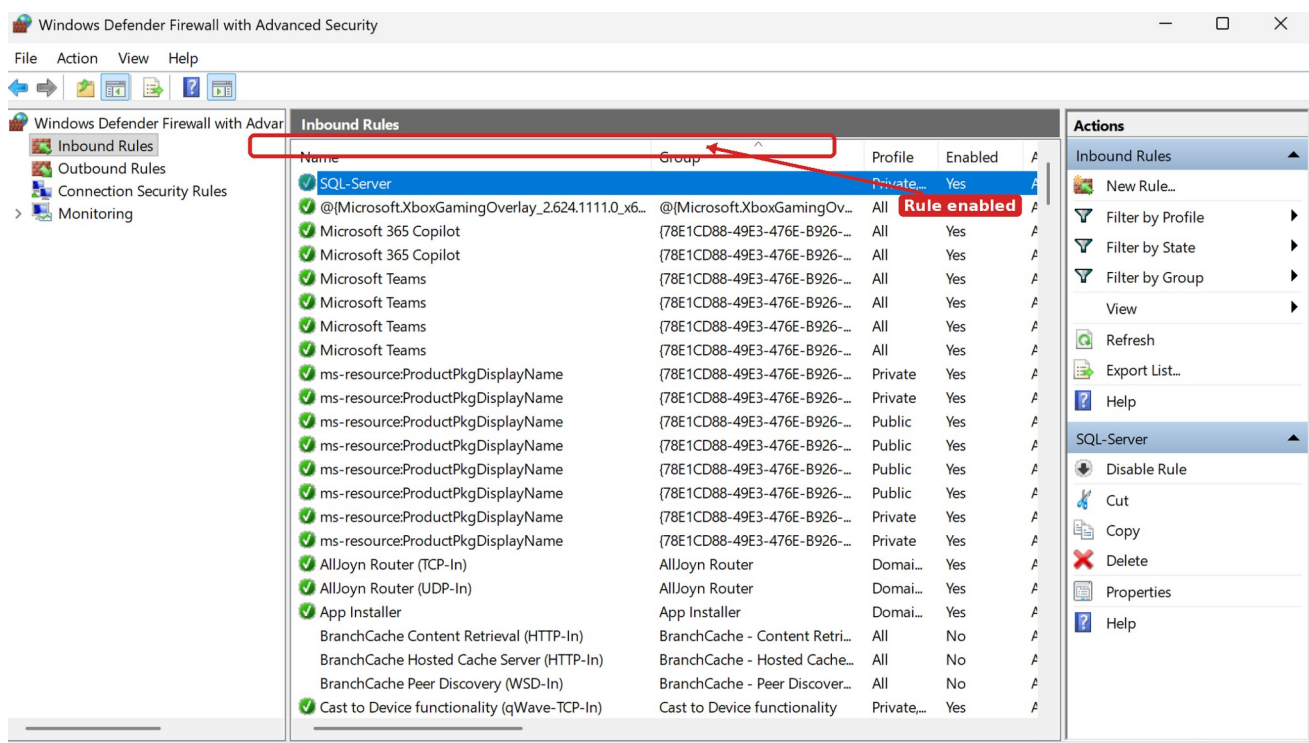


Figure 11 - The SQL-Server inbound rule is present and enabled.

7. Verify Mac -> VM TCP connectivity

Step 13 - Test port 1433 from macOS

Open Terminal on the Mac and run:

```
nc -vz <VM_IP> 1433
```

A successful result looks similar to:

```
Connection to <VM_IP> port 1433 [tcp/ms-sql-s] succeeded!
```

This is the key network checkpoint: If nc succeeds, VMware networking, SQL Server TCP listening, and the Windows firewall path are all working. Authentication can then be configured separately.

If nc fails

1. Re-run ipconfig and verify <VM_IP>.
2. Confirm VMware still uses Share with my Mac.
3. Confirm TCP/IP is enabled in SQL Server Configuration Manager.
4. Confirm SQL Server was restarted after the protocol/port change.
5. Confirm the SQL-Server inbound firewall rule is enabled and uses TCP 1433.
6. Confirm IPAll TCP Port is 1433 and TCP Dynamic Ports is empty.

8. Configure SQL authentication for macOS clients

Step 14 - Enable Mixed Mode

In SSMS, right-click the SQL Server -> Properties -> Security. Select SQL Server and Windows Authentication mode, click OK, then restart the SQL Server service.

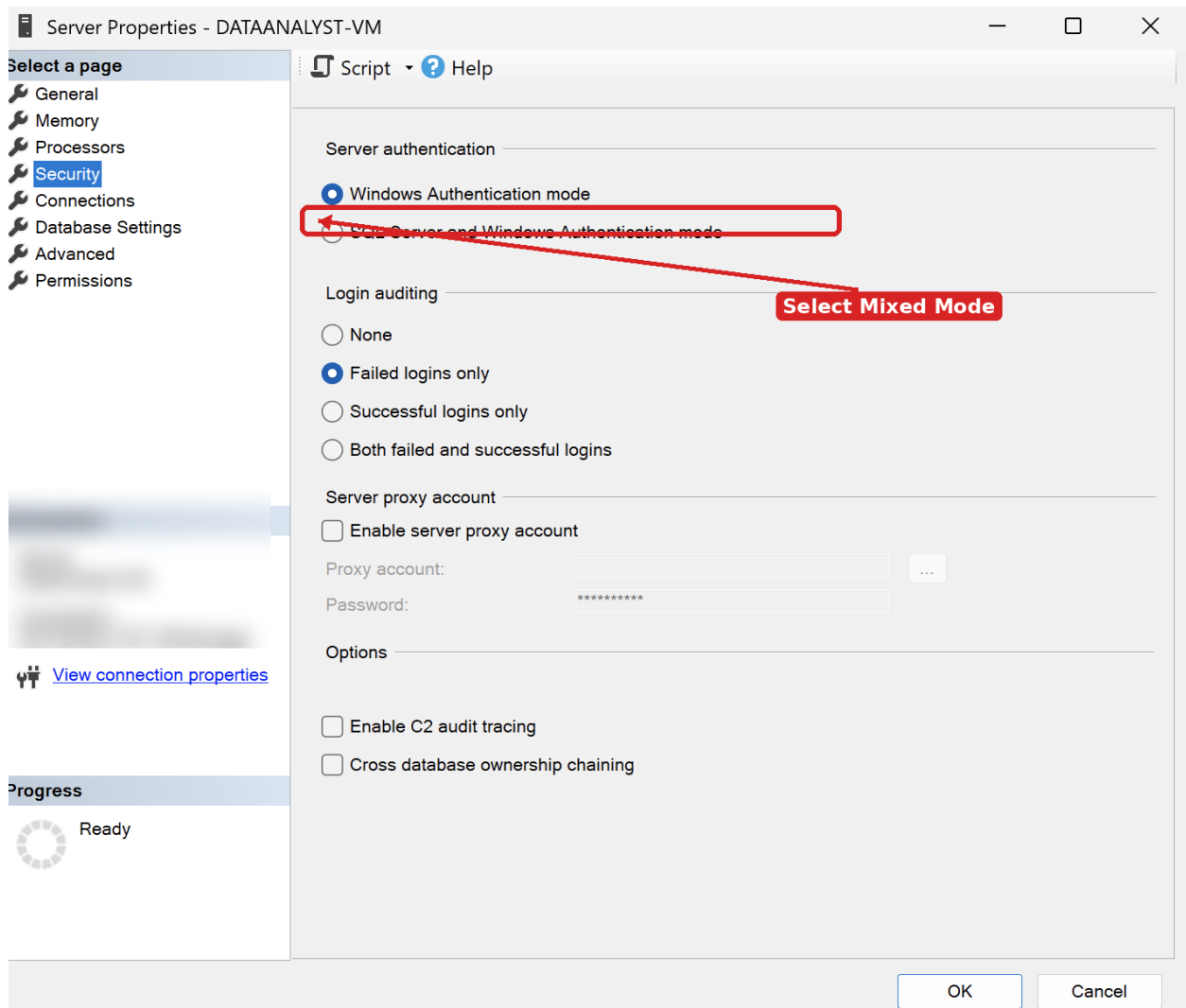


Figure 12 - Enable SQL Server and Windows Authentication mode (Mixed Mode). Server/user identifiers are blurred.

Step 15 - Use a SQL Login

The original verified setup enabled the sa login and confirmed that it could connect. That works for a private lab, but sa is a sysadmin account and should not be your normal development login.

Recommended for future use: Create a dedicated SQL Login with only the permissions required for your database. Use sa only when administrative access is actually necessary.

For example, in SSMS you can create a login under Security -> Logins -> New Login, choose SQL Server authentication, set a strong password, and map it only to the databases/roles it needs.

Authentication checkpoint

Before moving to VS Code, test the SQL Login inside SSMS. If the same username/password cannot log in from SSMS, VS Code will not be able to fix that problem.

9. Install the VS Code SQL Server extension

Step 16 - Install Microsoft SQL Server (mssql)

On the Mac, open VS Code -> Extensions. Search for SQL Server (mssql) and install the extension published by Microsoft.

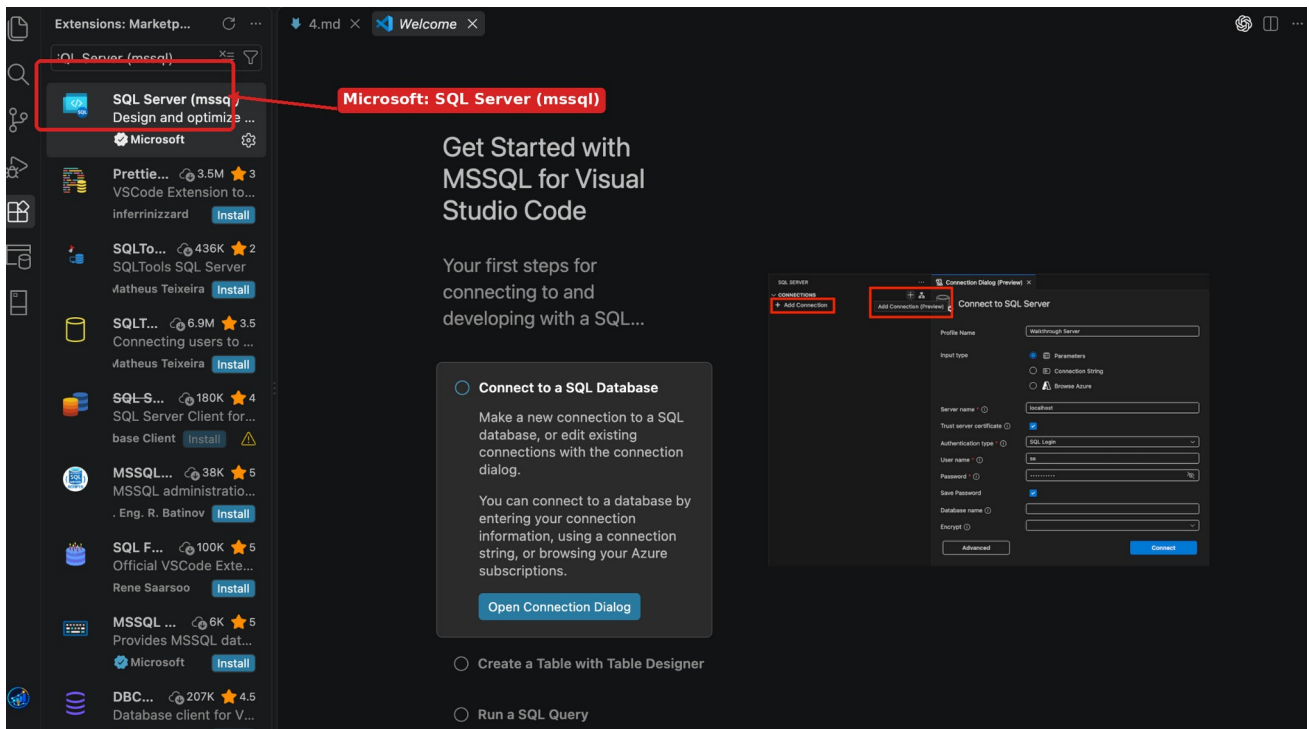


Figure 13 - Install SQL Server (mssql) by Microsoft.

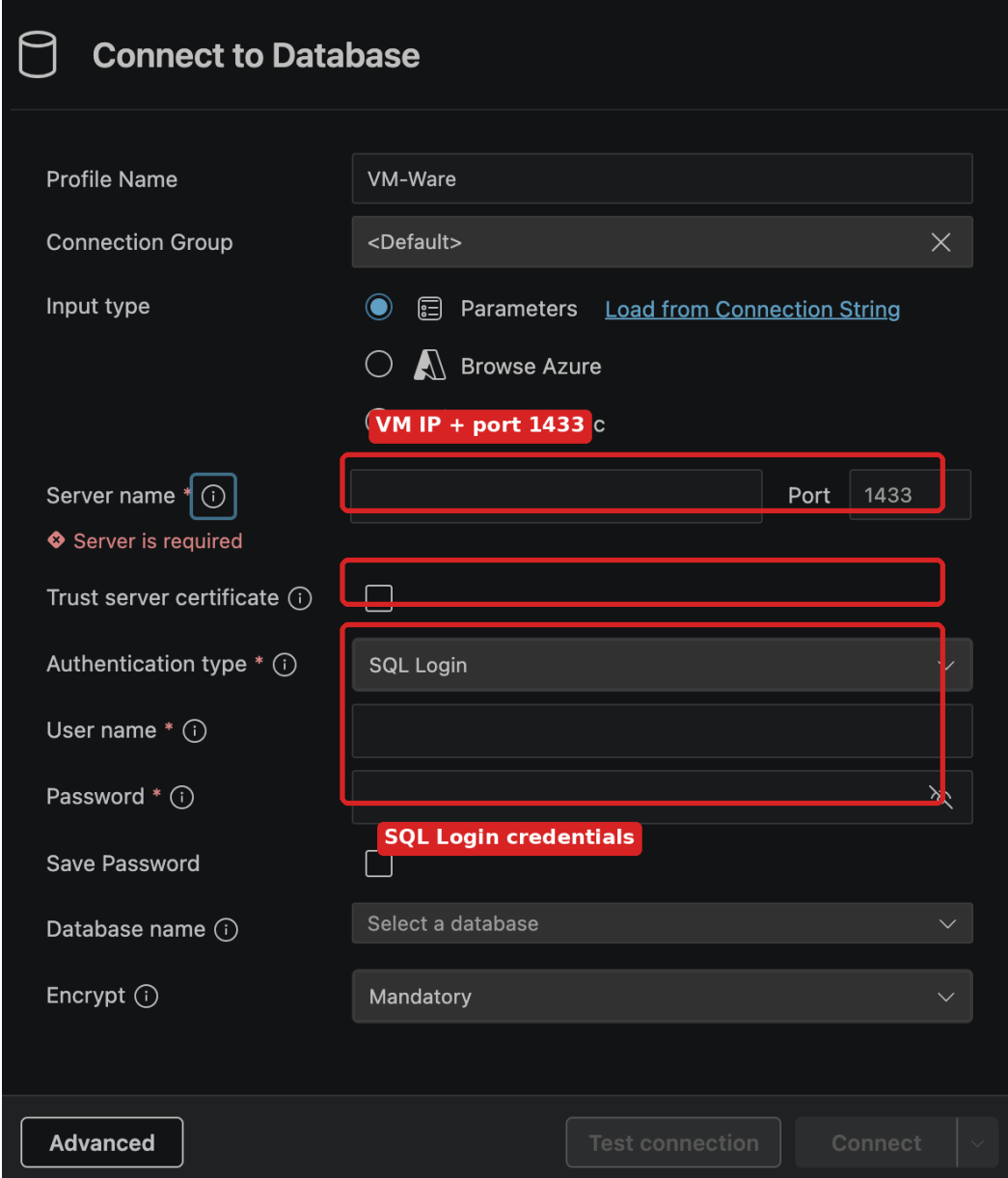
10. Create the VS Code connection

Step 17 - Open the connection dialog

Use the mssql extension welcome page or SQL Server sidebar and choose Open Connection Dialog / Add Connection.

Connection fields

Field	Value	Notes
Profile Name	Any friendly name	Example: Windows-VM-SQL
Server name	<VM_IP>	The IPv4 value from ipconfig
Port	1433	Must match SQL Server + firewall
Trust server certificate	Checked	Suitable for this local dev VM using a non-public/self-signed certificate
Authentication type	SQL Login	Requires Mixed Mode
User name	Your SQL login	Prefer a dedicated login; sa only for admin/lab use
Password	Your SQL password	Do not publish or hard-code it
Database name	Optional	Leave blank to connect first, or choose a database
Encrypt	Mandatory / default	Keep encryption enabled; trust the local dev certificate as above



The image shows the 'Connect to Database' dialog in VS Code. It has a dark theme. The title bar says 'Connect to Database' with a database icon. The form has several sections: Profile Name (VM-Ware), Connection Group (<Default>), Input type (Parameters selected, with a link to 'Load from Connection String' and an option for 'Browse Azure'), Server name (with a red box around the input field and a red label 'VM IP + port 1433' above it, and a Port field with '1433'), Trust server certificate (checkbox), Authentication type (SQL Login selected, with a red box around the dropdown and a red label 'SQL Login credentials' below it), User name (input field), Password (input field with a red box around it), Save Password (checkbox), Database name (dropdown with 'Select a database'), and Encrypt (Mandatory). At the bottom are buttons for 'Advanced', 'Test connection', and 'Connect'.

Connect to Database

Profile Name: VM-Ware

Connection Group: <Default>

Input type: ☒ Parameters [Load from Connection String](#) ☐ Browse Azure

Server name * Port: 1433

Trust server certificate ☐

Authentication type *

User name *

Password *

Save Password ☐

Database name

Encrypt

Advanced Test connection Connect

Figure 14 - VS Code mssql connection dialog. Use the VM IP, TCP 1433, SQL Login, and trust the local server certificate.

11. Test, connect, and run a query

Step 18 - Test the connection

Click Test connection first. If it succeeds, the network and SQL Login credentials are correct.

Step 19 - Connect

Click Connect. The server should appear in the SQL Server view / Object Explorer in VS Code.

Step 20 - Run a verification query

Create a new SQL query and run:

```
SELECT
    @@SERVERNAME AS ServerName,
    DB_NAME() AS CurrentDatabase,
    @@VERSION AS SqlServerVersion;
```

If results are returned, the full Mac -> VMware -> Windows -> SQL Server -> VS Code path is working.

Daily-use workflow after the one-time setup

7. Start the Windows VM.
8. Wait for SQL Server (MSSQLSERVER) to be running.
9. If needed, run ipconfig and confirm the VM address.
10. Open VS Code on the Mac.
11. Select the saved mssql connection profile and connect.
12. Open a .sql file and execute queries against the desired database.

Fast diagnostic: If VS Code suddenly cannot connect, first run `nc -vz <VM_IP> 1433`. If that fails, the issue is network/listening/firewall. If it succeeds, focus on SQL credentials, login status, database permissions, or certificate settings.

12. Troubleshooting matrix

Symptom	Likely area	What to check first
ping times out	Usually Windows ICMP firewall	Do not rely on ping. Test <code>nc -vz <VM_IP> 1433</code> .
nc to 1433 fails	Network / SQL TCP / firewall	VM IP, VMware NAT, TCP/IP enabled, restart, firewall rule, port 1433.
nc succeeds but VS Code test fails	Authentication / TLS / database	Mixed Mode, SQL Login/password, login enabled, Trust server certificate, database permissions.
SSMS localhost works but Mac cannot connect	Remote TCP path	Local SSMS may use Shared Memory. Enable TCP/IP and firewall 1433.
Worked yesterday, fails after restart	VM address changed	Run <code>ipconfig</code> again and update the VS Code server address.
Login failed for user	SQL authentication	Verify the same login in SSMS, Mixed Mode enabled, login not disabled, correct password.
Certificate trust error	Local SQL certificate	Keep Encrypt enabled and use Trust server certificate for this controlled local dev VM.
Connection timeout	Port not reachable / wrong IP	Run <code>nc</code> first; do not start by changing credentials.

A useful mental model

Troubleshoot in layers:

1. VM address -> 2. TCP port reachable -> 3. SQL authentication -> 4. Database permission -> 5. Query

Do not change multiple layers at once. The `nc` command is especially useful because it cleanly separates network/port problems from SQL authentication problems.

13. Security and maintenance recommendations

- Prefer a dedicated SQL Login instead of sa for daily development.
- Use a strong unique password and do not store it in source control.
- Keep SQL Server encryption enabled. Trusting the server certificate is acceptable for this controlled local VM, but production environments should use a properly trusted certificate.
- If desired, tighten the Windows Firewall rule Scope so only the VMware host/subnet can reach TCP 1433.
- Do not expose TCP 1433 to the public internet or configure router port forwarding for this local development scenario.
- After major VMware network changes, re-check the VM IP and the firewall profile classification.

14. One-page checklist for future setup

Check	Expected value
☐ VMware Network Adapter	Connect Network Adapter = ON; Share with my Mac = selected
☐ Windows VM address	Known from ipconfig
☐ SQL Server TCP/IP	Enabled
☐ SQL Server service	Restarted after TCP changes
☐ IPAll TCP Dynamic Ports	Empty
☐ IPAll TCP Port	1433
☐ Windows Firewall	Inbound TCP 1433 allowed for the VM's active profile
☐ Mac port test	nc -vz <VM_IP> 1433 succeeds
☐ SQL authentication	Mixed Mode enabled and SQL Login tested
☐ VS Code extension	SQL Server (mssql) by Microsoft installed
☐ VS Code server	<VM_IP>
☐ VS Code port	1433
☐ Trust server certificate	Enabled for this local dev VM
☐ Final validation	Test connection succeeds, then SELECT query returns results

Result: Once these checks pass, VS Code on macOS can query SQL Server running inside the VMware Windows VM without installing SQL Server directly on the Mac.